Cover Letter and Author Declaration

[Date]

To
The Organising/Editorial Committee
NIELIT Bio-AI Conclave 2026
NIELIT Guwahati, Assam, India

Subject: Submission of a full research paper, “What Near-Perfect Accuracy Does and Does Not Show on the UCI TCGA Pan-Cancer RNA-Seq Dataset: Leakage, Equivalence, Conformal Reliability and Data Efficiency”

Dear Members of the Committee,

I am pleased to submit the above full research paper for consideration at NIELIT Bio-AI Conclave 2026 under Track 2: Genomics, Multi-Omics & Precision Health. The work also relates to Track 3 (reproducibility of computational pipelines). [Confirm the primary track]

The paper examines the widely used UCI TCGA pan-cancer RNA-Seq dataset (801 tumours, five cancer types). With gene selection fitted inside every training fold and repeated cross-validation, seven classifiers reach 98.9–99.8% macro-F1, and no pair differs significantly. The paper quantifies when leaky gene selection inflates accuracy (strongly with permuted labels and small samples, barely with real labels at n = 801), and adds calibration, conformal prediction with abstention, and learning-curve analyses. It gives practical guidance on what near-perfect accuracy on this dataset can and cannot support; all code is released as Supplementary File S1.

Submitted files: manuscript (DOCX and identical PDF); supplementary document (DOCX and identical PDF); code archive (Supplementary File S1, ZIP) if the portal accepts it.

On behalf of all authors, I confirm that:

- the manuscript is original and has not been published previously in substantially the same form; [Confirm]
- the manuscript is not simultaneously submitted to or under consideration by another conference, journal or publication venue; [Confirm]
- all listed authors have reviewed and approved the final manuscript; [Confirm, and list co-authors if any]
- all authors have consented to the submission and agree to participate in the review and publication process;
- any actual or potential conflicts of interest have been disclosed in the manuscript; [Confirm the declaration]

- the work uses only publicly available datasets, which are cited, and no third-party copyrighted figures, tables or text requiring permission are reproduced;
- the use of generative AI and AI-assisted tools is declared in the manuscript; [Finalise the declaration]
- at least one author will complete the required BioAI Conclave 2026 author registration if the manuscript is accepted.

I will act as the corresponding author and remain available for communication until the review and finalisation process is complete.

Yours sincerely,

Er. Rishabh Aryan (Corresponding Author)
M.Tech (Artificial Intelligence and Data Science), Department of Computer Science and Engineering
Indian Institute of Information Technology Bhagalpur, Bhagalpur, Bihar, India
Email: rishabh.250201011@iiitbh.ac.in | ORCID: https://orcid.org/0009-0004-7595-9440

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